

What will happen If we relax the OLS assumptions?

Relaxing the ordinary least square assumptions^{OLS} can lead to various consequences, including multicollinearity, heteroscedasticity & autocorrelation. Here is a detailed explanation of each:

Multicollinearity:—

- Consequences: Multicollinearity occurs when two or more independent variables are highly correlated with each other. This can lead to unstable estimates of the regression coefficients, inflated variance & incorrect conclusions. For instance:

$$Y = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + u$$

⇒ If the independent variables X_1 & X_2 both affect each other then there will be multicollinearity among them.

Reasons: Multicollinearity can arise

due to various reasons, including:

- Overlapping variables (e.g., using GDP & GNP as independent variables)
- High correlation between variables (e.g., using both income & wealth as independent variables)
- Including irrelevant variables in the model.

Nature:

Multicollinearity is a problem of model specification, where the relationships between the independent variables are too strong.

How to remove:

- Drop one of the highly correlated variables.
- Use dimensionality reduction techniques (e.g., PCA, factor analysis).
- Use regularization techniques (e.g., LASSO, ridge regression).

Example:—

Suppose we have a linear regression model with two independent variables, X_1 & X_2 , and a dependent

variable Y . The data is as follows:

X_1	X_2	Y
1	2	3
2	4	6
3	6	9
4	8	12
5	10	15

In this example, X_1 & X_2 are highly correlated ($r=0.99$), which indicates multicollinearity. This can lead to unstable estimates of the regression coefficients.

Heteroscedasticity:

- **Consequences:** Heteroscedasticity occurs when the variance of the error term is not constant across all levels of the independent variables.

This can lead to inefficient estimates of the regression coefficients & incorrect conclusions.

- **Reasons:** Heteroscedasticity can arise due to various reasons, including:

- Non-linear relationships between the variables.
- Outliers or influential observations.
- Non-constant variance in the error term.

Nature: Heteroscedasticity is a problem of model specification, where the relationships between the variables are not properly captured.

Example:-

Suppose we have a linear regression model with one independent variable, X , and a dependent variable, Y . The data is as follows:

X	Y
1	2
2	4
3	6
4	10
5	20
6	40
7	80
8	160
9	320

⇒ In this example, the variance of the residuals increases as X increases, which indicates heteroscedasticity. This can lead to inefficient estimators of the regression coefficients.

How to Remove:

- Use transformations (e.g., log, square, root) to stabilize the variance.
- Use weighted least squares (WLS) to give more weight to observations with lower variance.
- Use robust standard errors (e.g., Huber white standard errors)

Autocorrelation:

- **Consequences:** Autocorrelation occurs when the error term is correlated with itself over time. This can lead to inefficient estimates of the regression coefficients & incorrect conclusions.

- **Reasons:** Autocorrelation can arise due to various reasons, including:

- Time-series data with strong temporal dependence.
- Omitted variables or incorrect model specification.
- Non-stationarity in the data.

Nature:

Autocorrelation is a problem of model specification, where the relationships between the variables are not properly captured over time.

Examples:-

Suppose we have a time series data with one dependent variable Y , & the following values:

Time	Y
1	10
2	12
3	14
4	16
5	18
6	20
7	22
8	24
9	26
10	28

⇒ In this example, the values of Y are highly correlated with each other over time, which indicates autocorrelation. This can lead to inefficient estimates of the regression coefficients & incorrect conclusions.

How To Remove:

- Use time-series models (e.g., ARIMA, SARIMA) to account for temporal

dependence.

- Use differencing or detrending to remove non-stationarity.
- Use robust standard errors (e.g., Newey-West standard errors) to account for autocorrelation.

